Neonatal jaundice

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Neonatal jaundice is a very common condition

Neonatal period = first 28 days of life

Outline of lecture

1. Physiology of NNJ
2. Other pathological causes of NNJ
3. NNJ hazard – KERNICTERUS
4. Conjugated hyperbilirubinemia
5. Clinical Approach to NNJ
6. Breastmilk jaundice

From Google

Clinical assessment

- Jaundice results from bilirubin accumulation in skin and sclera
  - yellow – unconjugated bilirubin
  - Yellow with green hue – conjugated bilirubin
- In neonates, jaundice is visible when it reaches 80-100 μmol/L (4.5-6.0 mg/dL)

How common?

- Over 50% of term infants and 80% of preterm infants will have clinical jaundice within the first few days of life
- More common in Asian population

Case

- Baby Amanda was born at 37 weeks, birth weight 2.8kg
  - No complications
  - Started breastfeeding
  - Appears mildly yellow from 48 hours of life
  - Weight loss of 8%, otherwise normal examination

Bilirubin is the breakdown product of hame

Physiology

1. Upstream
   - High Hb load in neonates (mean 17-19g/dL)
   - Short RBC lifespan (half life 80days)
2. Downstream
   - Immature liver metabolic function
     - Low ligandin (uptake of bilirubin into liver cells)
     - Low conjugation enzyme activity (UDGPT)

Heme metabolism

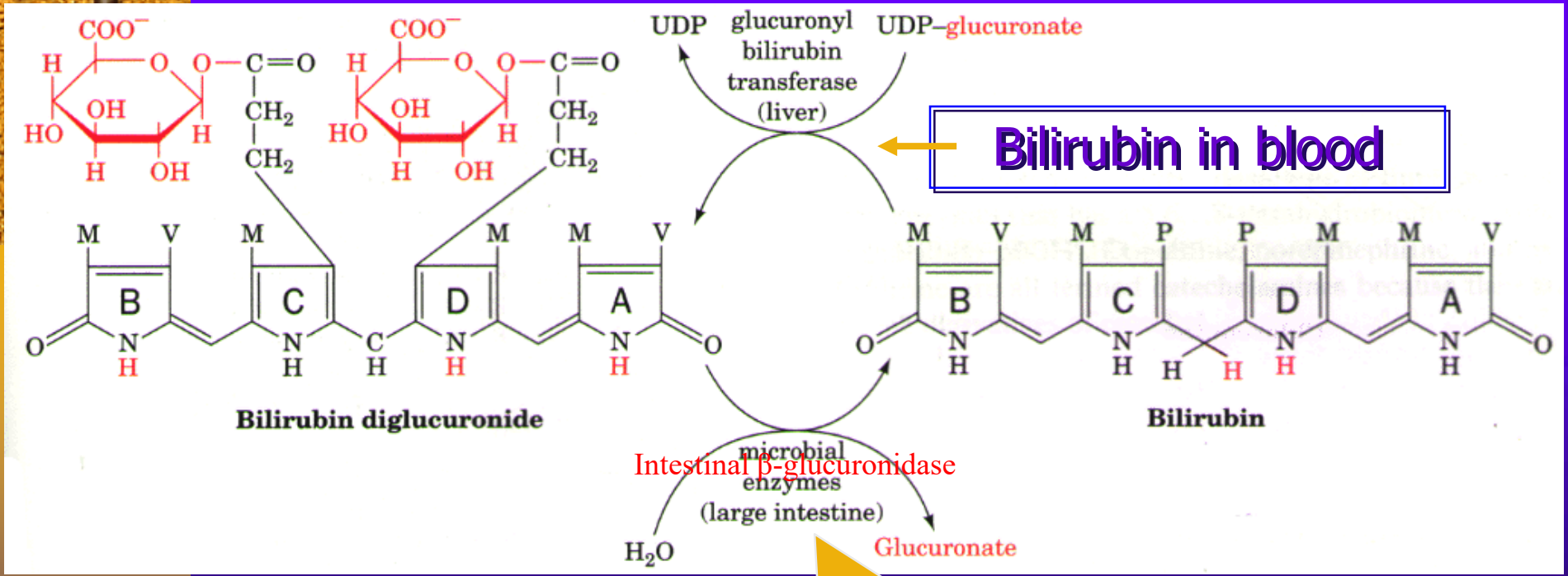
Reticuloendothelial system

hydrophobic

Heme → biliverdin → unconjugated bilirubin

binds to albumin and transport to liver

Link between unconjugated and conjugated bilirubin (= bilirubin diglucuronide)



Enterohepatic circulation

Physiology

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     - Low conjugation enzyme activity (UDGPT)
3. Enterohepatic circulation – "recycling" of bilirubin from gut to blood

- especially when feeding is minimal in the first 2-3 days of life

UDGPT

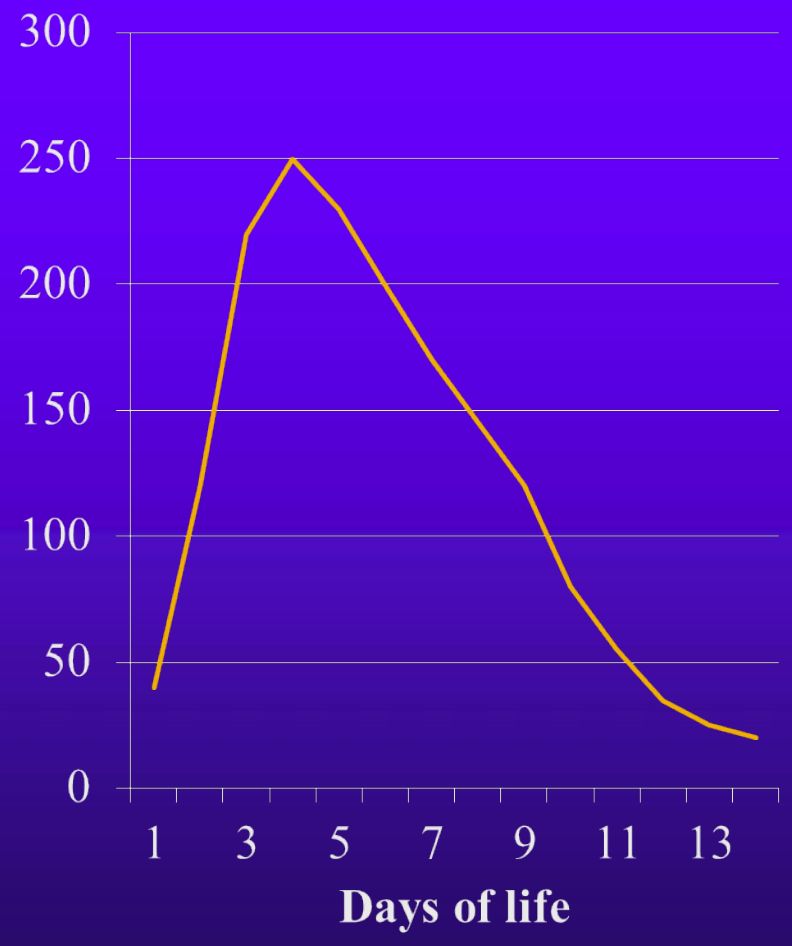
- Genetic variation (polymorphism) common → affects activity
  - Common: Gilbert syndrome
  - Uncommon: Criggler Najjar syndrome

Type of Hyperbilirubinemia in physiologic NNJ

- The accumulation of UNCONJUGATED bilirubin in blood

“Physiologic” jaundice – natural history

- apparent from day 2-3 of life
- peak at day 4-5
- subside by day 7-14
- Laboratory tests:
  - unconjugated hyperbilirubinemia (commonly >170 μmol/l)
  - normal hepatic ductal and parenchymal enzymes
  - absent conjugated fraction
  - Blood smear: no evidence of haemolysis



Hazards of NNJ

- Unconjugated bilirubin is hydrophobic

  → can cross lipid layers and blood brain barriers → causes damage to neural tissue

Luckily, most of the unconjugated bilirubin is bound to albumin

- Risk is higher when "free" bilirubin is high
  - When albumin is low
  - When there is competitive binding to albumin e.g. drugs

Hazards of NNJ

- KERNICTERUS
  - Death
  - Long term morbidity
    - dystonic cerebral palsy
    - hearing loss
    - upward gaze palsy

From Google

Risks of Kernicterus higher when

- Very high unconjugated bilirubin level in blood for a prolonged period
- High “free” bilirubin
- Hemolysis
- Sepsis
- Acidosis
- Hypoglycemia
- Prematurity

Causes of severe NNJ (unconjugated hyperbilirubinemia)

- Hemolysis
  - Immune causes e.g Rhesus incompatibility, ABO incompatibility
  - Non immune e.g. G6PD deficiency
  - Extravasation of blood e.g. cephalhematoma, intraventricular hemorrhage
  - polycythemia
- Inadequate feeding
- Sepsis

Clinical manifestation of bilirubin neuro-toxicity → **encephalopathy/kernicterus**

- Early: hypotonia, poor feeding, lethargy, hypertonia, opisthotonus, fever, convulsion, death

From Google

- Late: extrapyramidal signs, hearing loss

To reduce bilirubin toxicity

- Intervene before plasma bilirubin reaches "risky" level
  - term < 20mg/dl (340 μmol/l);
  - lower in preterm infants
- Avoid risk factors for kernicterus
- "free" bilirubin is not readily measurable

Clinical approach to NNJ (1)

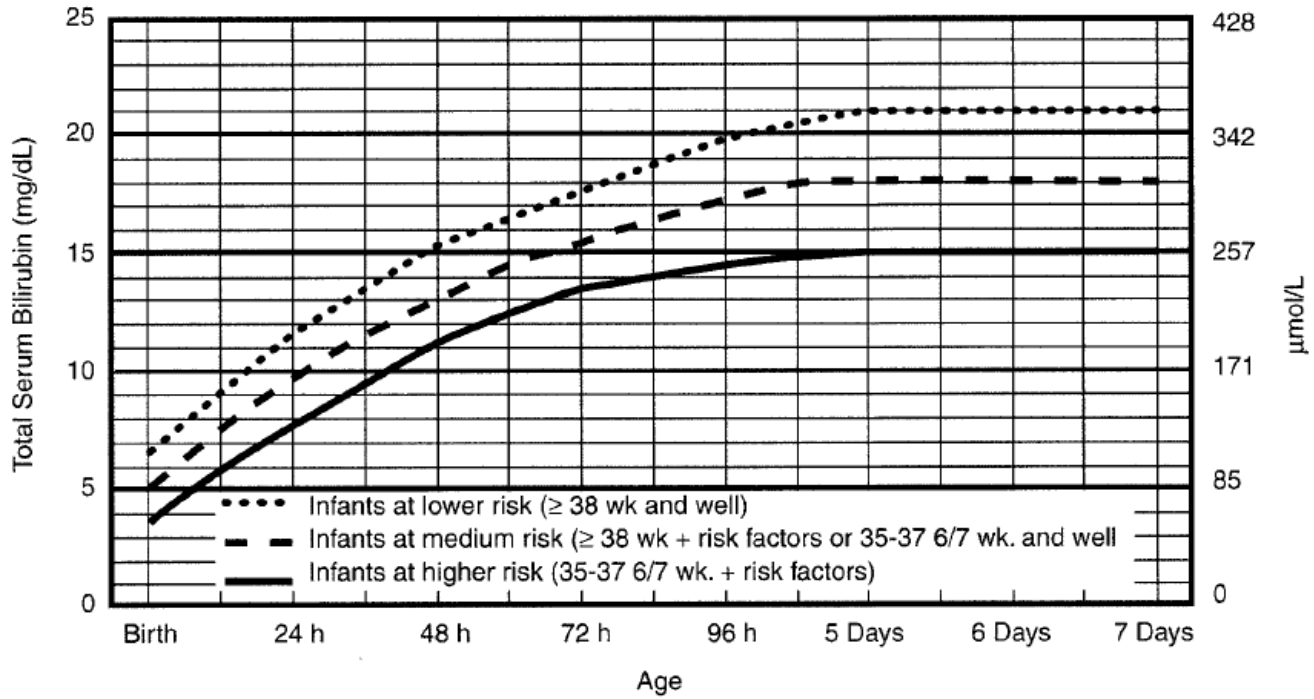
- History
  - Gestation, prenatal complications
  - Birth weight
  - Age of life
  - Feeding details
  - Blood groups/ G6PD status
  - Other postnatal complications e.g. infection

Clinical approach to NNJ (2)

- Physical examination
  - Severity of jaundice – using transcutaneous bilirubinometer
  - Hydration
  - Pallor/plethora
  - cephalhematoma/bruises
  - Neurologic state
  - Size of liver and spleen

Clinical approach to NNJ (3)

- Lab test: TOTAL SERUM BILIRUBIN



- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin.
- Risk factors = isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis, or albumin < 3.0g/dL (if measured)
- For well infants 35-37 6/7 wk can adjust TSB levels for intervention around the medium risk line. It is an option to intervene at lower TSB levels for infants closer to 35 wks and at higher TSB levels for those closer to 37 6/7 wk.
- It is an option to provide conventional phototherapy in hospital or at home at TSB levels 2-3 mg/dL (35-50mmol/L) below those shown but home phototherapy should not be used in any infant with risk factors.

Treatment of neonatal jaundice -- phototherapy

- Photo-isomerisation of bilirubin to less toxic (hydrophilic) and readily excretable forms (in bile and urine)
- wave length 460 nm (BLUE light spectrum)
- mild side effects:
  - corneal or retinal damage (use eye shield)
  - dehydration
  - skin rash
  - loose stool

From Google

Treatment of neonatal jaundice -- exchange transfusion

- Rapid way to remove plasma bilirubin by gradual exchange of patient's blood

- indicated when plasma bilirubin reaches a dangerous level (e.g. >380 μmol/l) or child already showing signs of neurotoxicity

Other possible treatment options

- IVIg for immune hemolytic jaundice

Breast milk jaundice 1

- Breast milk supply may be inadequate during first few days of life: decrease bilirubin excretion
  - increase enterohepatic circulation → reabsorption of unconjugated bilirubin (by intestinal β-glucuronidase deconjugation) into blood
  - THIS IS REFERRED TO AS BREASTFEEDING JAUNDICE (or NO BREASTFEEDING JAUNDICE)
- Certain ingredients in breast milk slow down conjugation (5-β-pregnane-3α,20- β-diol, nonesterified lung chain fatty acids, glucuronidase)
  - THIS IS REFERRED TO AS BREASTMILK JAUNDICE
- Evidence support genetic predisposition

Breast milk jaundice 2

This is a benign condition

- Jaundice may be prolonged (i.e. beyond 14 days of life) -- even up to 2-3 month old
- all unconjugated (do fractional bilirubin – direct and indirect)
- baby healthy otherwise
- Mothers should be advised to continue breast milk because of the other beneficial effects of breast milk

Summary 1

A newborn with clinical jaundice

1. History and physical exam.
   - ✓ ? compatible with natural course of "physiologic" jaundice
   - ✓ Any other identifiable factors
   - ✓ Any signs of neurotoxicity
2. Laboratory test → total serum bilirubin
3. Decide on treatment
4. Counsel parents on treatment and care of baby

How about conjugated hyper-bilirubinemia??

It does not cause kernicterus, but it can reflect sinister hepatobiliary diseases

Conjugated hyperbilirubinemia

- when direct bilirubin >2mg/dL (35μmol/L) or >15% of total serum bilirubin
- associated with
  - elevated parenchymal/ductal enzymes
  - deranged synthetic functions e.g. clotting factors, albumin
  - deranged detoxication functions e.g. hyperammonemia
  - deranged metabolic function e.g. hypoglycemia

Causes of conjugated hyperbilirubinemia: examples

Neonatal period

- Neonatal hepatitis (etiological agents mostly unknown)
- Biliary atresia / choledochal cyst
- Metabolic diseases e.g. citrin deficiency, galactosemia
- Parenteral nutrition associated cholestasis
- Syndromal disorders
- Neonatal hemochromatosis

Beyond neonatal period

- viral hepatitis (A-E)
- Drug induced hepatotoxicity (e.g. paracetamol)
- Wilson's disease
- Hemochromatosis, other metabolic diseases
- Others e.g. choledochal cyst, hereditary hyperbilirubinemia
- Cystic fibrosis

Obstructive jaundice – amendable to surgical treatment

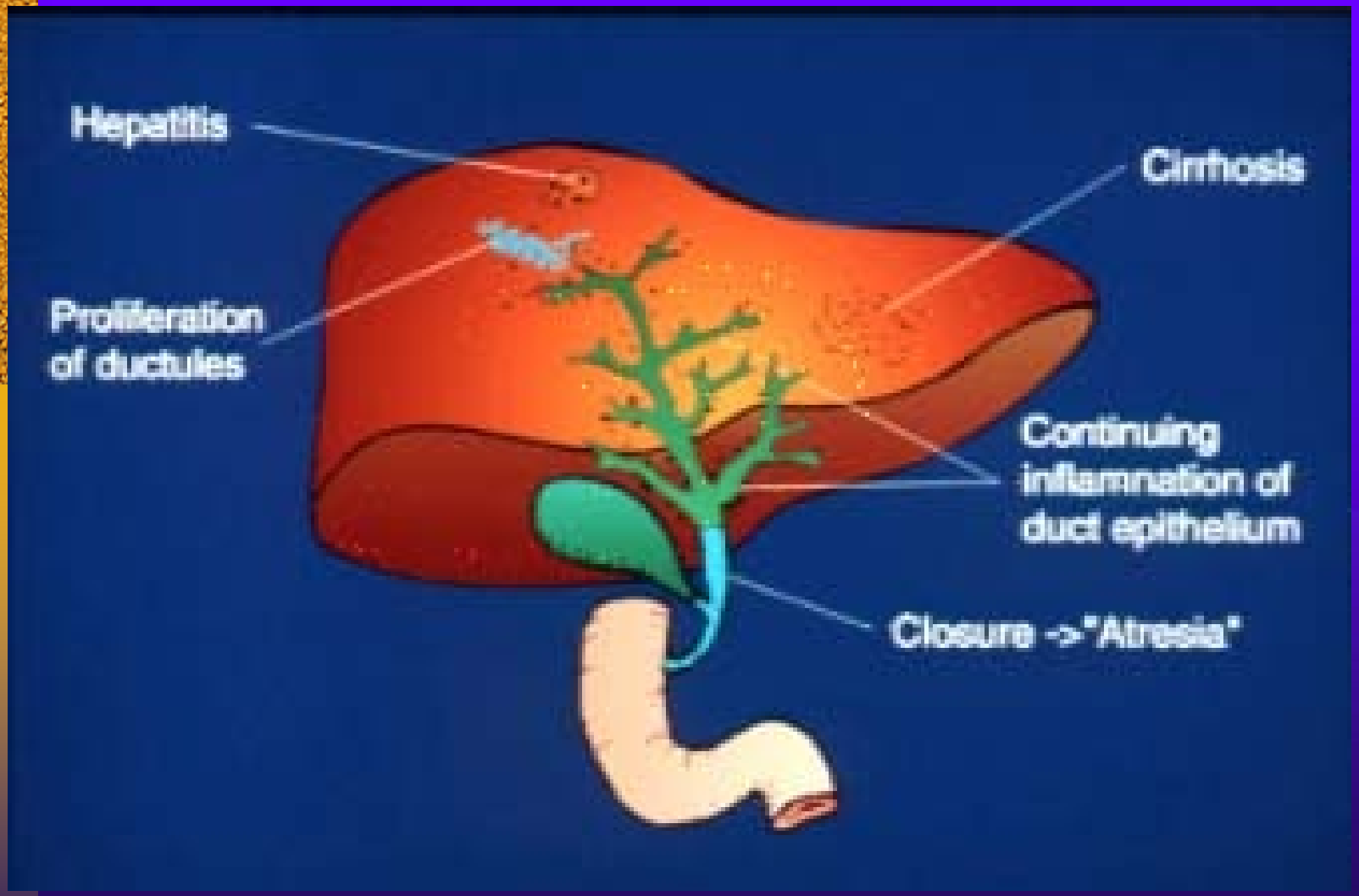
Conjugated hyperbilirubinemia

In neonatal period, RULE OUT

Biliary atresia

Choledochal cyst

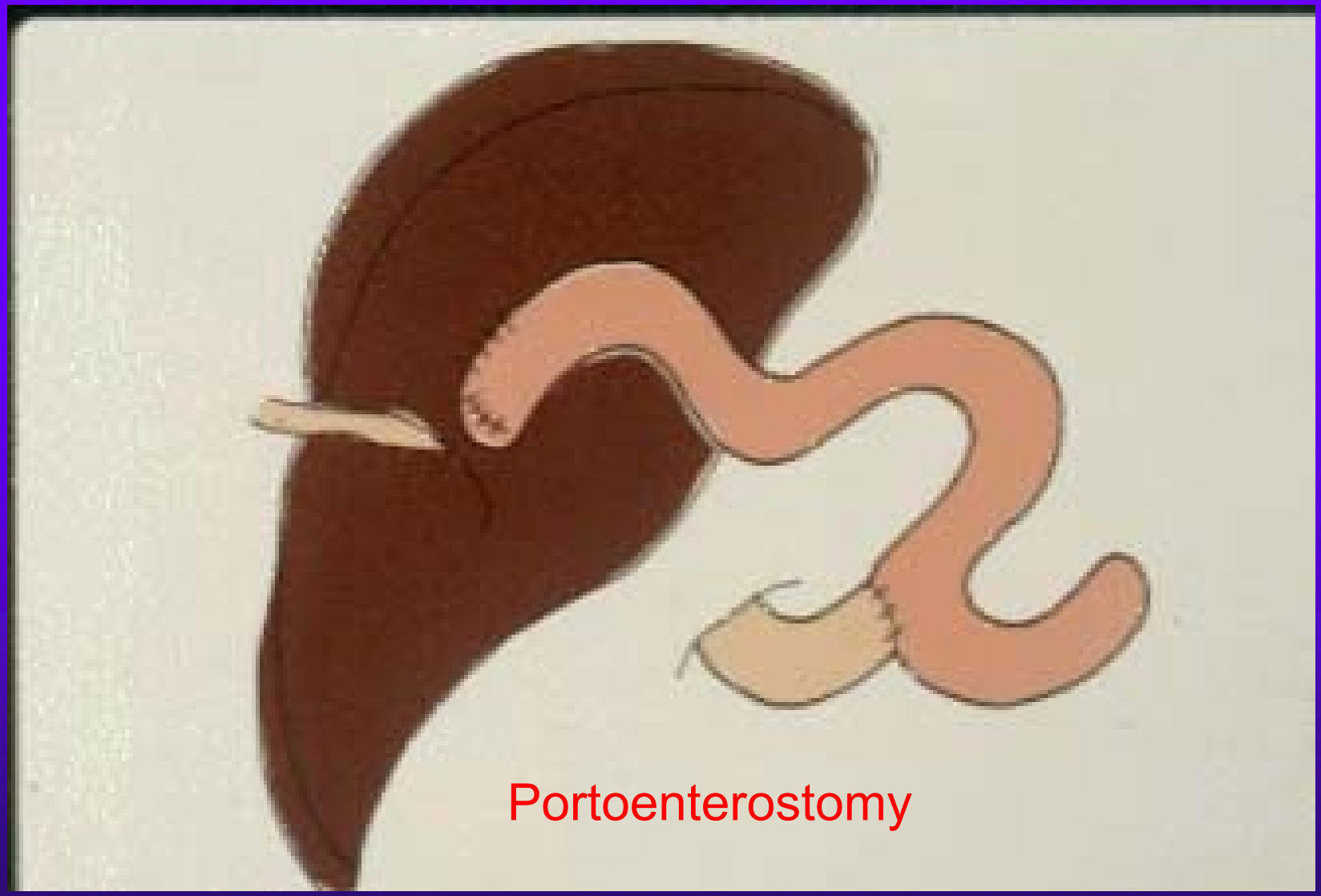
Biliary Atresia



Biliary atresia

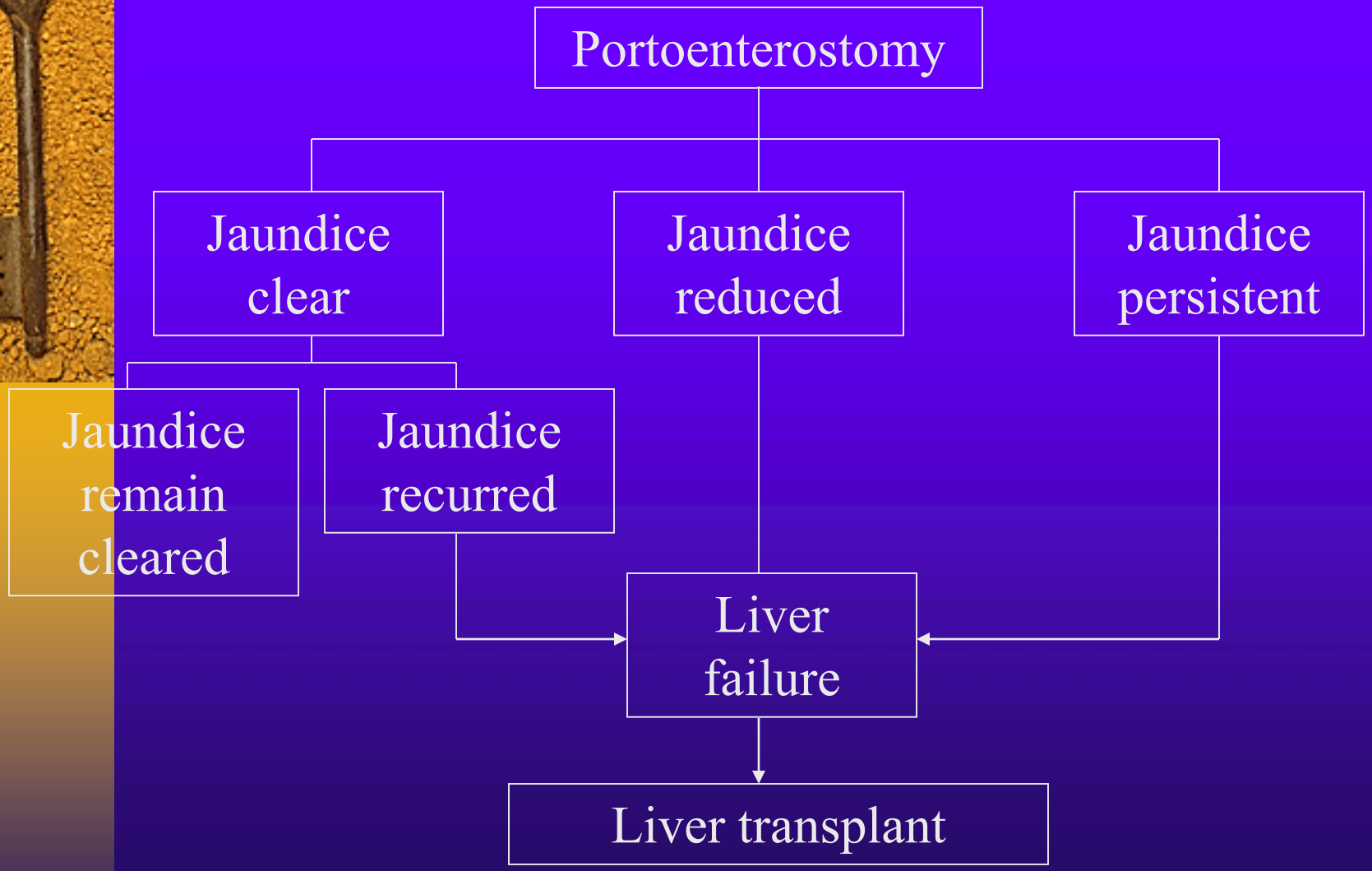
- Progressive obliterative cholangiopathy due to ductal obstruction
- Present during or beyond neonatal period with
  - prolonged jaundice
  - clay colour stool
  - tea colour urine
  - poor growth
  - Enlarged (or shrunken) liver and spleen
- Lab: Conjugated hyperbilirubinemia
- Progressive hepatic parenchymal damage and liver failure if untreated

- Portoenterostomy before 12 weeks → Best result
- Beyond 12 weeks →Poor result



Kasai operation

Management of Biliary Atresia



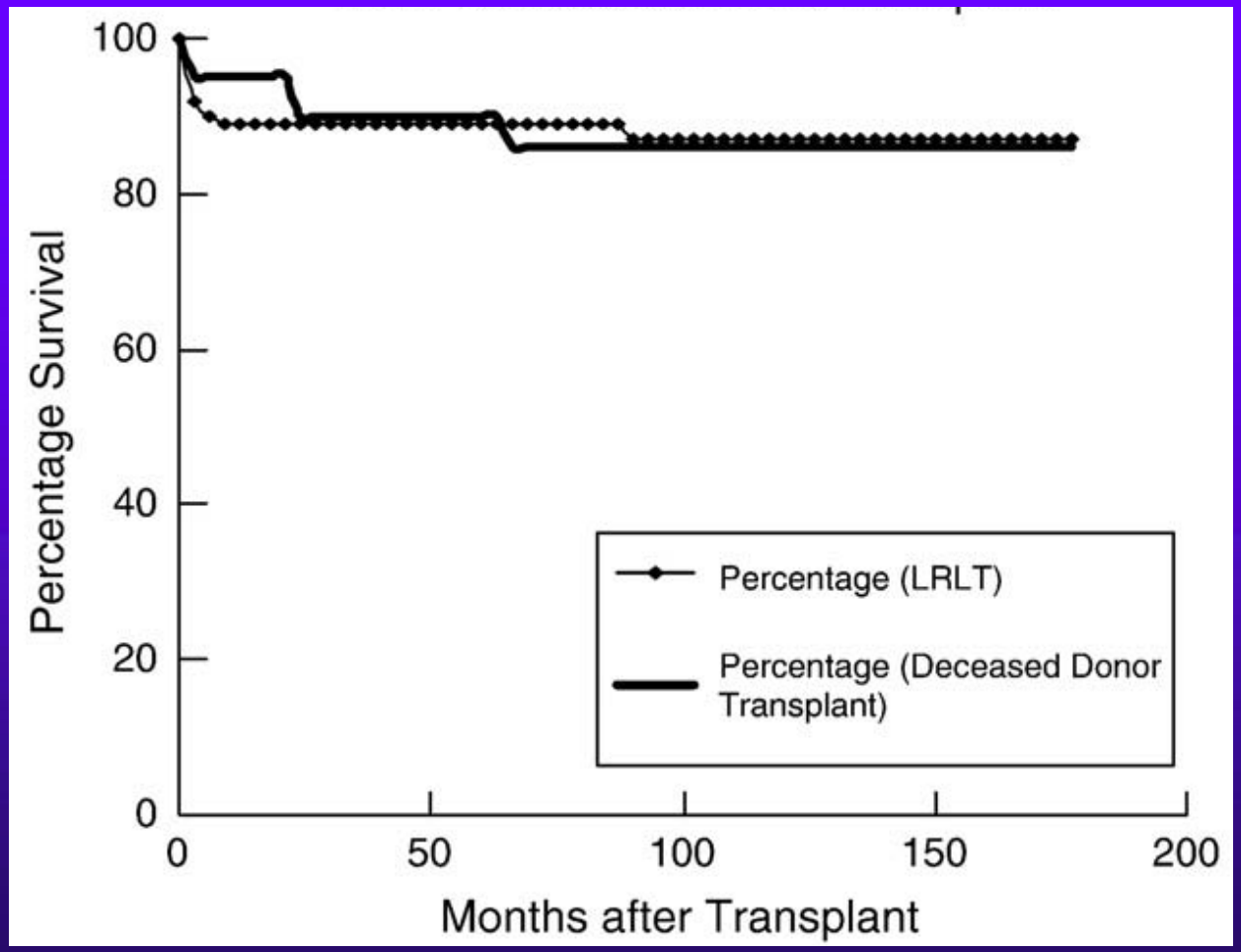
Indications for liver transplant referral

- Progressive liver failure
- Growth retardation
- Recurrent cholangitis
- Portal hypertension

The HKU experience

- Study period: 1993 to 2013
- Total number of recipients = 113
- LDLT: DDLT = 83: 30

LDLT vs DDLT (overall 10-year survival ~ 90%)



Chan et al. *J Paediatr Surg* 2010

Demographics (BA vs non BA)

	BA	Non- BA	P value
Number of patients	76	37	
Pre-LT PELD/MELD score	25.8 +/-7.2	8.2+/-4.6	0.001
Operative duration (mins)	512.6 +/- 48.1	470.0+/-52.3	0.4

Summary of LTx experience

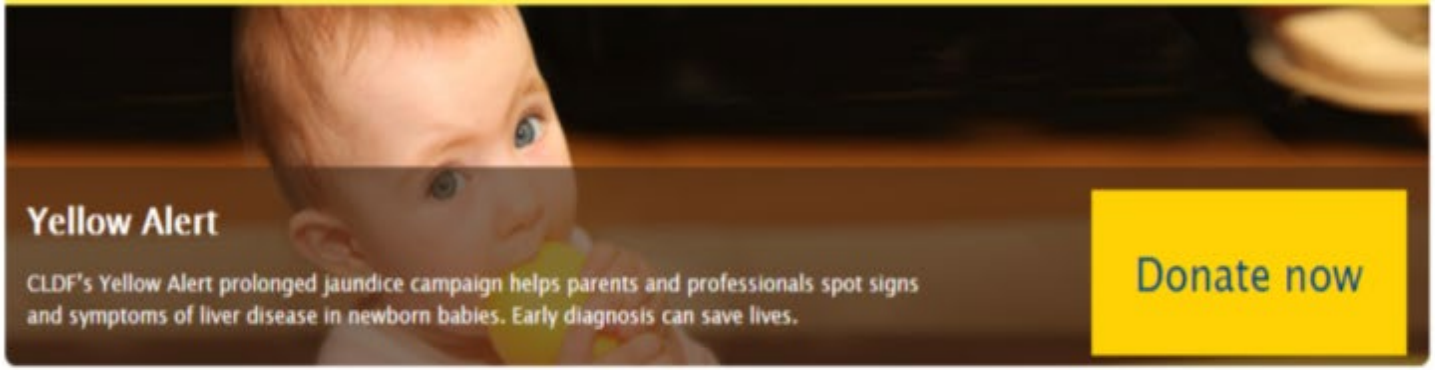
- BA remains the most common indication for paediatric liver transplant
- Liver transplant is an established treatment for patient who fail Kasai operation
- Results of LT for BA patients comparable to non-BA patients (~90% 5 to 10 year-survival)
- Timely referral can reduce morbidity and mortality

Yellow Alert

Keep up to date

Children's Liver Disease Foundation
fighting childhood liver disease

Home | Baby Jaundice | Breast Milk Jaundice | For Parents | For Professionals | NICE | Stories | FAQs | Glossary | Help us spread the word



Jaundice in newborn babies

Yellow Alert is CLDF's prolonged jaundice in newborn babies awareness campaign. Yellow Alert promotes the early diagnosis and appropriate referral for liver disease in newly born infants. It is vital that liver disease in newly born babies is identified and treated as early as possible.

- **Prolonged jaundice (yellowing of the skin and the whites of the eyes) can be a sign of liver disease.** It is defined as jaundice persisting beyond two weeks of age in a full-term baby and three weeks in a pre-term baby
- **Persistently pale coloured stools may indicate liver disease.** Healthy coloured stools can be described as English mustard yellow in bottle fed babies and daffodil yellow in breast fed babies
- **A baby's urine should be colourless.** Urine which is anything other than colourless could be a possible sign of liver disease

Babies with prolonged jaundice and/or persistently pale stools/urine which

Yellow Alert app

A new smartphone app to help identify the signs of serious liver disease in newborn babies.

Baby jaundice

Up to 80% of babies become jaundiced shortly after birth. Find out more about baby jaundice.

Your stories

Frequently asked questions

Take home message

- Neonatal jaundice
  - Physiology
  - Other causes of NNJ
  - Risk of severe NNJ
  - Clinical approach
  - Causes of conjugated hyperbilirubinemia in neonatal period
  - Biliary atresia

References

- http://emedicine.medscape.com/article/974786-overview
- American academy of paediatrics
- www.uptodate.com

Thank you

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!

JC
WCS

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and the others you may have missed.
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